Thermal radiation and color

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use peak wavelength as one piece of evidence about a thermal spectrum.

SUCCESS CRITERIA - I can...

1. Relate higher temperature to shorter peak wavelength.
2. Use kelvin in Wien's law.
3. Distinguish spectral peak from perceived color.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "blackbody" and "peak" to explain your model or result. Sentence starters are optional.

blackbody: an ideal absorber/emitter used in a thermal-radiation model

peak: the location of the largest value on a specified graph

kelvin: the absolute temperature scale used in thermal ratios

01 Warm-up: bring back an earlier idea

A light ray strikes a plane mirror at 20° from the normal. **Find** the reflection angle and label both angles on a sketch.

Word help: reflection = a wave returning from a boundary

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

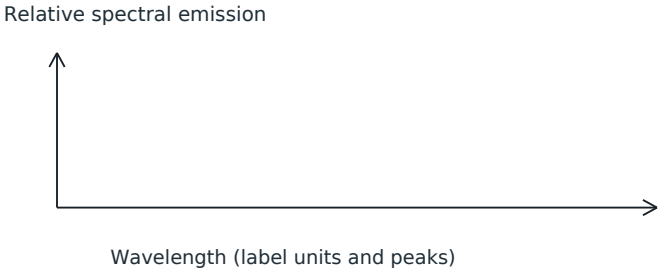
02 Worked example: explain the model

Reference: $\lambda_{max}\, T = 2.9\times10^{-3}$ m·K.
Ideal blackbody; T in kelvin, λ_{max} in meters. A shorter peak wavelength means higher temperature, not one exact visible color.

Worked reasoning: A thermal spectrum's peak shifts toward shorter wavelength as absolute temperature rises. Wien's relation $\lambda_{peak}\, T \approx 2.9\times10^{-3}$ m·K gives $\lambda_{peak} \approx 9.7$ micrometers at 300 K.

Your turn: Sketch two broad thermal spectra, labeling axes and the hotter peak's shift.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two ideal thermal emitters are at 3000 and 6000 K. Which has the shorter peak wavelength? Does that mean they emit only one color?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Relative spectral emission



Wavelength (label units and peaks)

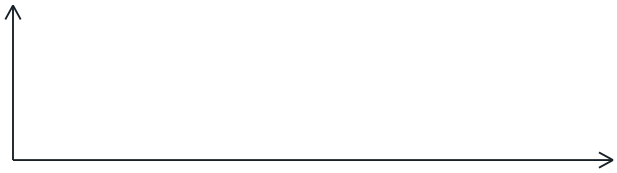
Marzano check - circle: 1 2 3

04 Apply the physics

Two thermal-emission models give peaks at about 970 nm for 3000 K and 480 nm for 6000 K.
Draw and label two broad spectra.
Compare peak wavelengths and identify the peak outside the approximate visible interval, 400-700 nm.

Sentence starter: Both … . They differ in … because … .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

05 Explain your reasoning

An object has an infrared peak but looks red-hot.
Explain how both can be true.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

An upward rope pulse is 4 cm high before reaching an ideal free end. Is the reflected displacement upright or inverted?
Sketch it.
Label a wave sketch to support the calculation or comparison.

Word help: ideal = using stated simplifying conditions

Sentence starter: The ... in my model represents ... . Its label shows ... .



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

Absolute temperature doubles. How does peak wavelength change?

Sentence starter: I notice ... . This supports ... because ... .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

08 STRETCH: test the model

If temperature doubles, compare peak wavelengths.
Explain why a room-temperature object can radiate even when it is not visibly glowing.

Sentence starter: Both ... . They differ in ... because ... .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3